TABLES

TABLE Ia

Physical Constants

Ionic Charge (Faraday) = 96,520 coulombs per gramme atom of univalent element.

Velocity of Light (in vacuo), $c = 2.99778 \times 10^{10}$ cm. sec.$^{-1}$.

Electronic Charge, $e = 4.803 \times 10^{-10}$ e.s.u. $= 1.602 \times 10^{-20}$ e.m.u. $= 1.602 \times 10^{-19}$ coulombs.

Rest Mass of Electron $m_0 = 9.107 \times 10^{-28}$ grammes.

Ratio of Charge to Mass of Electron, $e/m_0 = 1.759 \times 10^{8}$ coulombs per gramme.

Loschmidt's (Avogadro's) Number (Number of Molecules per Mole), $L = 6.024 \times 10^{23}$.

Planck's Quantum of Action (Planck's Constant), $h = 6.624 \times 10^{-27}$ ergs × sec. and $h = h/2\pi = 1.0543 \times 10^{-27}$ erg × sec.

Rydberg's Constant ($R = 2\pi^2 me^4/c\,h^3$) = 109,737 cm.$^{-1}$.

Mass of Electron = 5.486×10^{-4} a.m.u. (Atomic Mass Units).

Mass of Proton = 1.00758 a.m.u.

Mass of Hydrogen Atom = 1.00813 a.m.u.

Mass of Neutron = 1.00895 a.m.u.

Mass Ratio Hydrogen Atom to Electron, $M/m_0 = 1837.5$.

TABLE Ib

Physical Units

1 Million Electron Volts: 1 Mev $= 1.6 \times 10^{-6}$ erg $= 3{\cdot}83 \times 10^{-14}$ cal.

Energy Equivalent of Mass: 1 a.m.u. $= 1{\cdot}49 \times 10^{-3}$ erg.

Rest Energy of Electron, $m_0c^2 = 0{\cdot}51$ Mev $= 0{\cdot}8184 \times 10^{-6}$ erg.

Classical Electronic Radius, $r_e = e^2/mc^2 = 2{\cdot}82 \times 10^{-13}$ cm.

TABLE IC

Elementary Particles